Data-Driven Identification and MPC of Coupled Tank System using Bilinear Koopman Realizations and Physics-Informed Neural Networks

Amir Vanegas, Julio Barón-Velandia , Nelson Leonardo Díaz-Aldana and Duvan Tellez-Castro

***Abstract*— Real-time control of multivariable nonlinear processes requires models balancing high fidelity with computational tractability. This paper compares two data-driven paradigms: Bilinear Koopman Realizations and Physics-Informed Neural Networks. While standard Koopman approaches seek global linearization, we leverage a bilinear framework in the lifted functional space to preserve the natural coupling of control-affine systems. Simultaneously, PINNs ensure physical consistency by embedding conservation laws into the learning objective. To facilitate high-performance control, both surrogate models are integrated into a nonlinear model predictive control scheme using the CasADi framework, enabling efficient algorithmic differentiation for optimization. Simulation results for a quadruple tank system demonstrate that the Bilinear Koopman model offers superior tracking accuracy and robustness compared to the PINN-based approach. We conclude that the Bilinear Koopman realization, by avoiding the noise sensitivity observed in neural architectures, provides a more reliable, structure-aware solution for the identification and control of complex hydraulic benchmarks.**

***Index Terms*— Predictive Control, Neural Networks, Nonlinear Optimization, Nonlinear Systems, System Identification.**

I. INTRODUCTION

Real-time control of complex nonlinear processes requires models that balance accuracy with computational efficiency [1]. Traditional first-principles modeling often incurs high computational costs in online estimation and Model Predictive Control due to the recursive solution of differential equations [1]- [4]. To address these limitations, data-driven frameworks have emerged as a flexible alternative, capturing complex nonlinearities without relying on local linearization [3], [6]. Modern approaches, such as Bilinear Koopman Realizations and Physics-Informed Neural Networks, offer a scalable path for system identification by leveraging frameworks adaptable to increasing system dimensions. These methodologies provide a high-fidelity alternative that bypasses the extensive analytical redesign typically required by classical control techniques as systems become more complex [4], [6].

Within this context, the Koopman Operator provides a mathematical foundation for analyzing nonlinear dynamics by acting as an infinite-dimensional linear operator in the observables space [3], [5]. While traditional Extended Dynamic Mode Decomposition (EDMD) seeks linear lifting, recent advances emphasize Bilinear Koopman Realizations. As argued in [15], every control-affine system admits a bilinear realization in the lifted space, even when finite-dimensional linear representations fail. This framework enables structured control while preserving natural state-input coupling [2], [3]. However, despite their algebraic elegance, these purely data-driven approximations can struggle without enforced physical consistency, particularly under measurement uncertainty.

To mitigate these generalization and noise sensitivity issues, PINNs integrate differential equations as constraints into the loss function [7]. This architecture enables accurate system identification even with sparse datasets, as the model is forced to adhere to governing physical principles [4], [11]. The resulting synergy between deep learning and physics-based regularization enhances robustness against noise, overcoming the limitations of traditional black-box machine learning while maintaining the flexibility of data-driven methods [4], [12].

However, the question of whether these physics-informed architectures can outperform operator-theoretic methods in control-affine scenarios remains an open area of investigation. We bridge this gap by developing a data-driven framework that leverages Bilinear Koopman Realizations to preserve the natural coupling of nonlinear systems, providing a structured alternative to neural approximations. Specifically, this work contributes a rigorous comparative analysis of Bilinear Koopman and PINN-based modeling tailored for the identification and predictive control of the nonlinear quadruple tank system originally proposed in [8]. By embedding these surrogate models into a Nonlinear Model Predictive Control (NMPC) scheme using the CasADi framework, we quantify the trade-offs between structure-aware operators and physics-regularized learning. Our study evaluates both approaches using simulation-generated data to assess tracking accuracy and computational reliability, ultimately offering a selection criterion for real-time control in complex hydraulic benchmarks characterized by cross-coupled dynamics and square-root nonlinearities [2], [8].

The remainder of the paper is organized as follows: Section II defines the problem and the mathematical models. Section III outlines the implementation and NMPC formulation. Section IV discusses simulation results and performance metrics, while Section V provides concluding remarks.

II. PRELIMINARIES

This section formalizes MPC optimization, Koopman Operator, Bilinear Koopman Realizations and PINN architectures, establishing the mathematical foundation for system identification and control.

All authors are with Faculty of Engineering, Universidad Distrital Francisco José de Caldas, Bogotá D.C., Colombia {azvanegasc,jbaron,nldiazaduatellezc,} @udistrital.edu.co

A. Model Predictive Control

MPC computes optimal control actions by solving a constrained optimization problem over a finite time horizon at each sampling instant k [2]. Unlike traditional control laws, MPC leverages a surrogate model to predict the future evolution of the system, allowing the controller to anticipate complex dynamics and counteract disturbances before they affect the process.

The framework guides the system by minimizing a cost functional J over a prediction horizon N, which balances tracking performance, actuator effort, and smoothness

$$\min_{\mathbf{u}_0,\ldots,\mathbf{u}_{N-1}} J = \sum_{i=0}^{N-1} \left(\|\hat{\mathbf{h}}_{k+i+1} - \mathbf{r}_{k+i+1}\|_{\mathbf{Q}}^2 + \|\mathbf{u}_i\|_{\mathbf{R}}^2 + \|\Delta\mathbf{u}_i\|_{\mathbf{R}_d}^2 \right).$$

This optimization is subject to the system dynamics and physical operational limits:

$$\begin{aligned} \hat{\mathbf{x}}_{i+1} &= f_{\text{model}}(\hat{\mathbf{x}}_i, \mathbf{u}_i), \quad \hat{\mathbf{x}}_0 = \mathbf{x}_k, \\ \mathbf{u}_{\min} &\le \mathbf{u}_i \le \mathbf{u}_{\max}, \\ \Delta\mathbf{u}_{\min} &\le \mathbf{u}_i - \mathbf{u}_{i-1} \le \Delta\mathbf{u}_{\max}, \end{aligned}$$

Where $\hat{\mathbf{h}}$ represents the predicted tank levels and $\mathbf{r}$ is the desired reference. The matrix $\mathbf{Q}$ defines the importance of tracking precision, $\mathbf{R}$ represents the penalty on energy consumption or control effort, and $\mathbf{R}_d$ accounts for the desire for temporal smoothness in the control signal. Since the controller's success depends entirely on the fidelity of the mapping f_{model}, this study evaluates the integration of Bilinear Koopman Realizations and Physics-Informed Neural Networks as the predictive engines within this optimization framework.

B. Koopman Operator Theory

Koopman operator theory provides a mathematical foundation for analyzing nonlinear dynamics by shifting the perspective from state-space trajectories to the evolution of functions. It describes nonlinear systems through an infinite-dimensional linear operator $\mathcal{K}$ that acts on a space of scalar-valued observables g, such that $\mathcal{K}g = g \circ \mathbf{f}$ [6]. In practice, this framework seeks to find a finite-dimensional *lifting* mapping $\mathbf{z} = \psi(\mathbf{h})$ that embeds the nonlinear manifold into a higher-dimensional space where the dynamics appear linear. However, while this linear lifting is mathematically elegant for autonomous systems, the introduction of external control inputs often breaks the global linearity, necessitating a more flexible representation to capture state-input interactions.

C. Bilinear Koopman Realizations

To bridge the gap between global linearization and control-affine systems, recent advances favor Bilinear Koopman Realizations over traditional linear approximations. As established in [15], every control-affine system admits a bilinear representation in the lifted functional space, even when a strictly linear one is unattainable. This approach approximates the dynamics by allowing the control input $\mathbf{u}$ to act multiplicatively with the lifted states $\mathbf{z}$

$$\mathbf{z}_{k+1} = \mathbf{A}\mathbf{z}_k + \sum_{j=1}^{m} (\mathbf{B}_j \mathbf{z}_k) u_{j,k} + \mathbf{D}\mathbf{u}_k.$$

Within this structure, $\mathbf{A}$ captures the autonomous drift, $\mathbf{B}_j$ represents state-input interactions, and $\mathbf{D}$ characterizes direct additive actuator effects. To recover physical variables, we identify an output matrix $\mathbf{C}$ via Ridge regression to satisfy $\hat{\mathbf{h}}_k = \mathbf{C}\mathbf{z}_k$. While this decomposition significantly improves prediction fidelity, the potential lack of physical consistency in purely data-driven mappings motivates the integration of physics-based constraints through neural architectures.

D. Physics-Informed Neural Networks

PINNs approximate dynamical system solutions by embedding governing physical equations directly into the network's optimization objective. The framework trains a neural network $\hat{\mathbf{h}}(t) = \mathcal{N}(t, \mathbf{v}; \boldsymbol{\theta})$ to represent system trajectories while strictly adhering to physical laws [4], [8]. A composite loss function $\mathcal{L}(\boldsymbol{\theta})$ governs the training process

$$\mathcal{L}(\boldsymbol{\theta}) = w_d \mathcal{L}_{data} + w_p \mathcal{L}_{phys} + w_{ic} \mathcal{L}_{ic}, \tag{1}$$

where weighting coefficients w_i balance each term's contribution. Specifically, $\mathcal{L}_{data}$ quantifies the mean squared error against measured data, while $\mathcal{L}_{ic}$ enforces initial conditions. To maintain physical consistency, the residual term $\mathcal{L}_{phys}$ leverages automatic differentiation to evaluate the governing law:

$$\mathbf{r}(t) = \frac{d}{dt}\hat{\mathbf{h}}(t) - \mathbf{f}(\hat{\mathbf{h}}(t), \mathbf{v}(t); \boldsymbol{\omega}).$$

By minimizing the norm of this residual over collocation points, the network converges toward a solution that satisfies fundamental physics principles. This approach effectively bridges the gap between purely data-driven methods and first-principles modeling.

III. PROBLEM FORMULATION

This section establishes the mathematical foundation of the quadruple tank system and formalizes the data-driven identification and control objectives addressed in this study.

A. Quadruple Tank System Dynamics

The quadruple-tank process serves as a multivariable (MIMO) benchmark for control, characterized by complex cross-coupled dynamics. As Fig. 1 illustrates, the setup utilizes two pumps to distribute flow among four interconnected tanks through a specific valve configuration. Mass conservation and Bernoulli's principle define the following nonlinear state-space representation:

$$\begin{aligned} \dot{h}_1 &= -\frac{a_1}{A_1}\sqrt{2gh_1} + \frac{a_3}{A_1}\sqrt{2gh_3} + \frac{\gamma_1 k_1}{A_1} v_1, \\ \dot{h}_2 &= -\frac{a_2}{A_2}\sqrt{2gh_2} + \frac{a_4}{A_2}\sqrt{2gh_4} + \frac{\gamma_2 k_2}{A_2} v_2, \\ \dot{h}_3 &= -\frac{a_3}{A_3}\sqrt{2gh_3} + \frac{(1-\gamma_2)k_2}{A_3} v_2, \\ \dot{h}_4 &= -\frac{a_4}{A_4}\sqrt{2gh_4} + \frac{(1-\gamma_1)k_1}{A_4} v_1, \end{aligned}$$

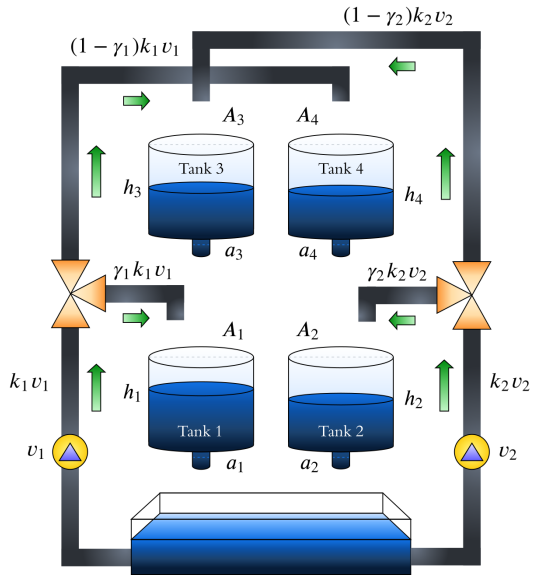

Fig. 1. Quadruple Tank System Representation.

where h_i represents the water level in each tank and v_i denotes the pump voltages. The parameters A_i and a_i correspond to the cross-sectional and outlet areas, respectively; k_i are the flow constants, and $\gamma_1, \gamma_2 \in [0,1]$ represent the valve settings that determine the flow distribution. The system output vector $\mathbf{y}$ integrates the measured levels of all four tanks:

$$\mathbf{y} = [h_1, h_2, h_3, h_4]^\top.$$

Torricelli's Law introduces square-root nonlinearities, while the system equations reveal strong cross-coupling; specifically, the levels h_3 and h_4 directly drive the dynamics of the lower tanks. These structural interdependencies demand high-fidelity identification to achieve effective regulation within the control framework [8].

B. Identification and Control Objectives

This work primarily aims to identify accurate surrogate models for the quadruple tank system using two distinct data-driven paradigms: Bilinear Koopman Realizations and PINNs. These models must capture nonlinear interactions and cross-coupling effects to enable precise trajectory tracking. Consequently, the control problem involves designing an MPC strategy to regulate the lower tank levels (h_1, h_2) by manipulating pump voltages $\mathbf{v} = [v_1, v_2]^\top$.

IV. METHODOLOGY

This section details the data-driven identification strategy and the MPC framework designed to regulate the quadruple tank's nonlinear cross-coupled dynamics.

A. Bilinear Koopman Identification

The identification process implements the transition function f_{model} by mapping tank levels $\mathbf{h} \in \mathbb{R}^4$ into a higher-dimensional feature space. To capture Bernoulli-driven nonlinearities and trigonometric couplings, we employ the dictionary $\mathbf{z} = \boldsymbol{\psi}(\mathbf{h}) = [\mathbf{h}, \sqrt{|\mathbf{h}|}, \sin(\mathbf{h}), \cos(\mathbf{h}), \mathbf{h}^2]^\top$. To identify the operator matrices, we construct a regression matrix $\mathbf{X}_{\text{reg}}$ containing the lifted states $\mathbf{z}_k$, the control inputs $\mathbf{u}_k$, and their bilinear interactions $\mathbf{z}_k u_{j,k}$. The weights for the dynamical evolution $\mathbf{W}_{\text{dyn}} = [\mathbf{A}, \mathbf{B}_1, \ldots, \mathbf{B}_m, \mathbf{D}]$ are estimated via Ridge regression

$$\min_{\mathbf{W}_{\text{dyn}}} \|\mathbf{Z}_{\text{next}} - \mathbf{W}_{\text{dyn}}\mathbf{X}_{\text{reg}}\|_F^2 + \alpha_1 \|\mathbf{W}_{\text{dyn}}\|_F^2,$$

where α_1 ensures numerical stability against sensor noise and F denote the Frobenius norm. Subsequently, we identify the output matrix $\mathbf{C}$ by solving a separate regularized least-squares problem

$$\min_{\mathbf{C}} \|\mathbf{H} - \mathbf{C}\mathbf{Z}\|_F^2 + \alpha_2 \|\mathbf{C}\|_F^2$$

using the full dataset. This two-step identification allows the NMPC to evolve the system state within the lifted manifold while maintaining a precise, high-fidelity projection back to the physical level measurements.

B. PINN Identification

As an alternative, a PINN identifier approximates the function f_{model} through a deep residual network $\mathcal{N}$. The architecture processes the input vector $[\mathbf{x}_k, \mathbf{u}_k] \in \mathbb{R}^6$ to predict the subsequent state $\hat{\mathbf{x}}_{k+1}$. To enforce physical consistency, the training process minimizes the composite loss in (1), where the residual term $\mathcal{L}_{res}$ penalizes deviations from the discrete-time mass balance

$$\mathbf{R}_k = \hat{\mathbf{x}}_{k+1} - \left(\mathbf{x}_k + \Delta t \cdot \mathbf{f}_{phys}(\mathbf{x}_k, \mathbf{u}_k)\right).$$

To maintain numerical stability, f_{model} utilizes a scaling mapping based on the training extrema $[h_{\min}, h_{\max}]$. This normalization, in conjunction with optimization clipping, restricts the predicted trajectories $\hat{\mathbf{x}}$ to the feasible hydraulic range during MPC execution.

C. MPC Approach

The proposed framework utilizes a NMPC strategy centered on the CasADi suite. By leveraging algorithmic differentiation, the controller transforms the surrogate models f_{model} into differentiable expressions for high-performance gradient-based solvers.

The Bilinear Koopman model structures the optimization as a Quadratic Programming problem solved via OSQP, exploiting the model's linearity in the lifted space. Conversely, the PINN-based controller manages neural nonlinearities through Sequential Quadratic Programming. Both implementations enforce real-time feasibility using a warm-start strategy from $\mathbf{u}^*_{k-1}$. This unified approach ensures that performance differences stem strictly from the predictive fidelity of the underlying data-driven paradigms rather than the optimization algorithms.

V. CASE STUDY

This section details the implemented environment, the training protocol for neural surrogates, and the physical parameters of the quadruple tank benchmark.

A. System Parameters

Plant simulation and predictive models rely on the geometric and flow constants from the Johansson benchmark [9], summarized in Table I. These values define minimum phase characteristics and the multivariable coupling strength of the system.

TABLE I
PHYSICAL PARAMETERS OF THE QUADRUPLE TANK SYSTEM.

Symbol	Description	Value	Unit
$A_{1,3}$	Area (Tanks 1 & 3)	28.00	cm^2
$A_{2,4}$	Area (Tanks 2 & 4)	32.00	cm^2
$a_{1,3}$	Outlet area (Tanks 1 & 3)	0.071	cm^2
$a_{2,4}$	Outlet area (Tanks 2 & 4)	0.057	cm^2
k_1, k_2	Pump flow constants	3.33, 3.35	cm^3/Vs
γ_1, γ_2	Flow split ratios	0.70, 0.60	–
g	Gravity	981.00	cm/s^2

B. Data Acquisition and Excitation

To achieve persistent excitation, data collection spans 5000 s with a sampling period $dt = 0.05$ s. A 4th-order Runge-Kutta method (RK45) integrates the nonlinear plant from initial conditions $\mathbf{h}_0 = [12.4, 12.7, 1.8, 1.4]^\top$ cm.

Excitation signals $\mathbf{u} = [v_1, v_2]^\top$ consist of a Pseudo-Random Binary Sequence (PRBS) within $[0, 6]$ V, switching every 50 s. Reproducibility is ensured via a Mersenne Twister (MT19937) generator with a fixed seed of 100. Furthermore, to emulate realistic conditions, additive white Gaussian noise (AWGN) with $\sigma = 0.1$ affects both control inputs and output measurements.

C. Identification Parameters and Training

The framework implements the PINN using Python and *PyTorch*, accelerating operations via an *NVIDIA GeForce GTX 1050 Mobile* GPU and *Intel Core i5-8300H* CPU. Both models adopt the configurations in Table II. Before training, a Kalman filter with covariance matrices $\mathbf{Q} = 10^{-5}\mathbf{I}$ and $\mathbf{R} = 10^{-2}\mathbf{I}$ processes the raw simulation data to mitigate measurement noise. While the Bilinear Koopman model operates directly on these filtered scales to preserve structural mapping, the PINN framework subsequently normalizes the measurements to $[0, 1]$ to ensure numerical stability and faster convergence. For the neural architecture, training utilizes a dual-phase strategy: an Adam optimizer performs the global search, followed by L-BFGS for second-order refinement of physical residuals.

TABLE II
PARAMETERS FOR KOOPMAN AND PINN IDENTIFICATION

Koopman Parameter	Value	PINN Parameter	Value
Observables (n_{obs})	20	Hidden Layers	4
Regularization (λ)	0.01	Neurons per Layer	48
Snapshot count (N)	5000	Activation Function	SiLU
Time step (Δt)	0.05 s	Adam Epochs	4000
$\mathbf{K}$ matrix size	4×20	L-BFGS Max Iter.	4000

D. Identification Results

Multi-step prediction performance evaluation compares Bilinear Koopman Realization and PINN trajectories against a State-space linearization, a standard Linear Koopman model, and actual tank levels. Fig. 2 illustrates this comparison over a validation window (2500–5000 s). Both the Bilinear Koopman model and the PINN achieve exceptional fitting, capturing the system nonlinearities with high structural fidelity and minimal deviation from the ground truth. While the State-space and Linear Koopman approximations provide a reasonable baseline, they exhibit a noticeable drift in highly nonlinear regions.

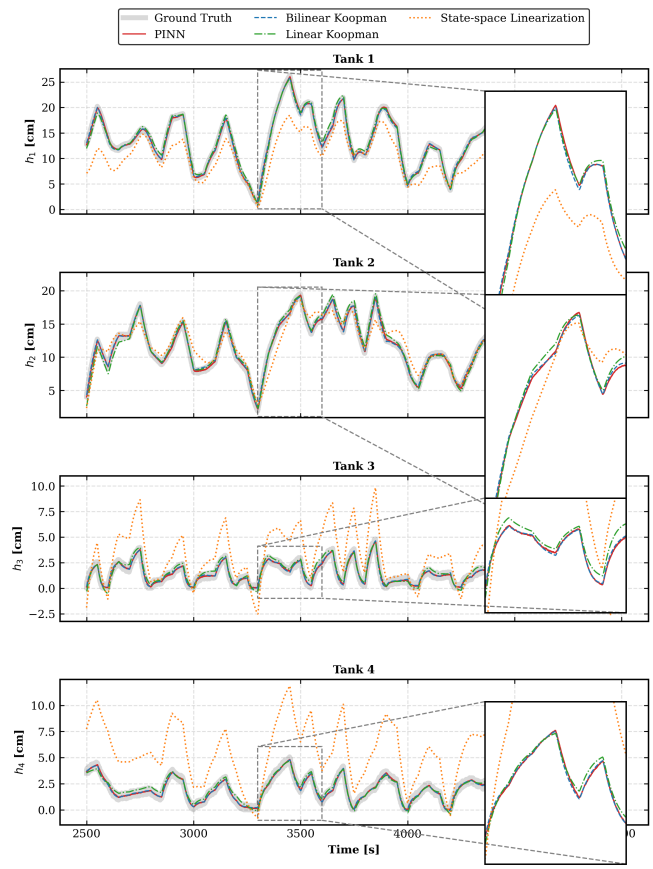

Fig. 2. Identification results: Bilinear Koopman vs Linear Koopman vs PINN vs State-space Linearization.

E. Model Validation Metrics

Quantitative assessment reveals that both Bilinear Koopman and PINN architectures achieve exceptional fidelity, consistently reaching VAF values above 99% across all tanks. As shown in Table III, these models significantly outperform Linear Koopman and State-space Linearization baselines, which exhibit substantial drift and total failure (0% VAF) in the upper stages. This gap underscores the necessity of nonlinear lifting or physical regularization to capture complex hydraulic couplings.

While both methods demonstrate near-identical accuracy, the Bilinear Koopman realization slightly leads in h_3 and

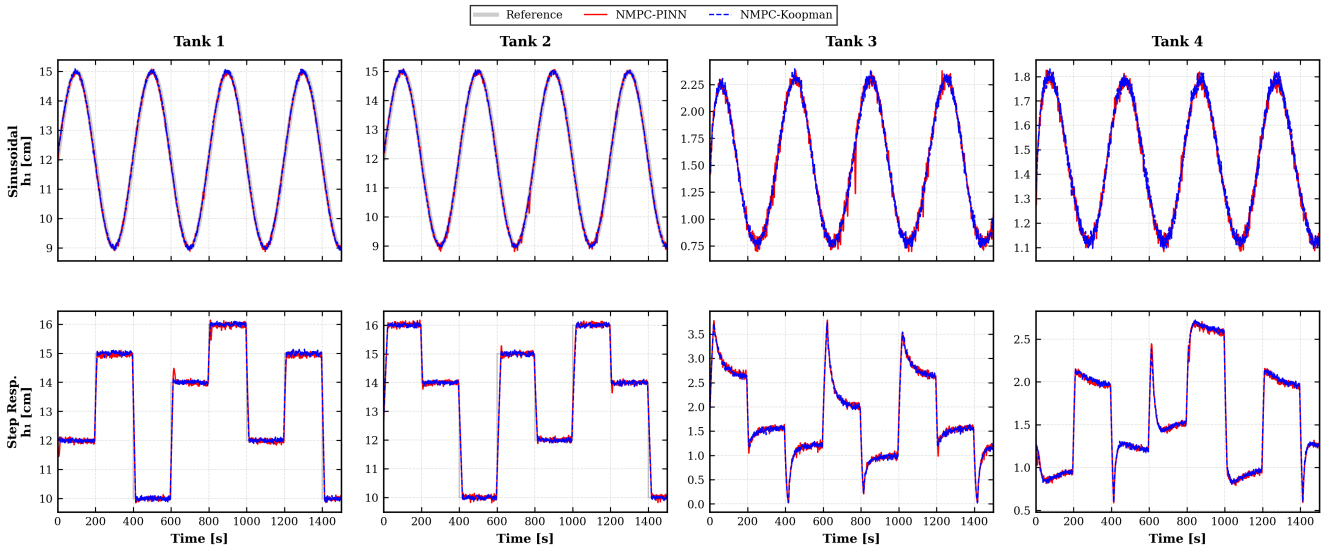

Fig. 3. Control performance comparison between NMPC-Bilinear Koopman and MPC-PINN: Sinusoidal Tracking and Step Response Cascade.

h_4 by exploiting the natural control-affine structure. These results confirm that embedding system knowledge—through bilinear lifting or physics-informed loss—provides a robust representation superior to traditional local approximations.

TABLE III
DETAILED PREDICTION METRICS PER TANK.

Model	Tank	RMSE (cm)	VAF (%)
PINN	h_1	0.1996	**99.86**
	h_2	**0.1714**	**99.84**
	h_3	0.1190	99.12
	h_4	0.1090	99.09
Bilinear Koopman	h_1	**0.1993**	**99.86**
	h_2	0.1871	99.81
	h_3	**0.1160**	**99.16**
	h_4	**0.1068**	**99.12**
Linear Koopman	h_1	0.7129	98.23
	h_2	0.4735	98.83
	h_3	0.3366	93.01
	h_4	0.2555	95.05
State-space Linearization	h_1	3.4272	77.28
	h_2	2.1867	74.86
	h_3	2.7749	0.00
	h_4	3.8656	0.00

F. Controller Real-Time Execution and Feedback Loop

The NMPC operates in a closed-loop with a sampling period of $\Delta t = 1$ s. At each step, the controller receives filtered measurements and computes the optimal sequence using a warm-start strategy from $\mathbf{u}^*_{k-1}$. While the Bilinear Koopman model enables near-instantaneous QP solutions, the PINN-based approach maintains a constant execution frequency via CasADi. Robustness is verified by injecting AWGN into sensors ($\sigma = 0.1$) and actuators ($\sigma = 0.15$), evaluating resilience to stochastic disturbances.

Table IV summarizes the NMPC hyperparameters used for both predictive models. The weighting matrices prioritize tracking accuracy in the lower tanks ($\mathbf{Q}$) while ensuring control effort economy ($\mathbf{R}$) and actuator smoothness ($\mathbf{R}_d$).

TABLE IV
NMPC HYPERPARAMETERS.

Parameter	Value	Description
N_p	25	Prediction horizon
N_c	1	Control horizon
$\Delta\mathbf{u}_{\max}$	0.5 V	Slew-rate limit
Weighting Matrices		
$q_{1,2}$	500	Tracking weights (h_1, h_2)
$q_{3,4}$	0	Stability weights (h_3, h_4)
$\mathbf{R}$	$0.01 \cdot \mathbf{I}$	Control effort weight
$\mathbf{R}_d$	$1.0 \cdot \mathbf{I}$	Slew-rate (smoothness) weight

G. Control Scenarios and Tracking Performance

Closed-loop simulations run for 1500 s with a sampling period of $\Delta t = 0.2$ s to evaluate the agility and robustness of the proposed predictors. The first scenario employs a Step Response Cascade where h_1 and h_2 alternate between 12, 15, 10, 14 and 16 cm every 200 s to quantify transient handling and settling time. Subsequently, a MIMO Sinusoidal Tracking scenario stresses the multivariable couplings through the reference $\mathbf{r}(t) = 12.0 + 3.0\sin(2\pi t/400)$, assessing how accurately the data-driven models capture the nonlinear oscillations of the quadruple tank system.

H. Control Performance Analysis

Fig. 3 compares the closed-loop tracking of both architectures. While both NMPC schemes maintain high fidelity despite multivariable coupling, differences emerge in transient behavior. Notably, the Bilinear Koopman NMPC exhibits faster convergence and settling times than its PINN-based counterpart. This stability stems from the Koopman model's ability to capture the system structure without learning high-frequency sensor noise. Consequently, the Koopman

realization ensures smoother transitions within the $[0,6]$ V manifold, preventing the oscillatory artifacts observed in the more sensitive neural approach.

I. Control Validation Metrics

Table V summarizes the global error analysis for both evaluated scenarios. In sinusoidal tracking, the Bilinear Koopman NMPC achieves a superior RMSE of 0.0961 cm, surpassing the PINN-based approach. While both models perform competitively under smooth trajectories, the Koopman realization demonstrates a more precise alignment with the reference by leveraging its structured bilinear form.

TABLE V
GLOBAL PERFORMANCE METRICS FOR MPC COMPARISON.

Scenario	Metric	Bilinear Koopman	PINN	Unit
Sinusoidal Tracking	RMSE	**0.0961**	0.1049	cm
	SSE	**138.59**	165.14	cm^2
	NRMSE	**1.60**	1.75	%
	RTF	**46.97**	8.89	-
Step Response Cascade	RMSE	**0.9470**	1.2753	cm
	SSE	**7174.84**	13011.63	cm^2
	NRMSE	**15.78**	21.26	%
	RTF	**48.94**	9.00	-

The Bilinear Koopman model yields a lower NRMSE, RMSE and SSE than the PINN, as the latter introduces oscillatory transients by overfitting high-frequency sensor noise during training. This sensitivity stems from the PINN's tendency to overfit localized stochastic fluctuations during the training phase, where the loss function may prioritize residual minimization over global dynamical smoothness. In contrast, the Koopman-based controller effectively rejects disturbances, ensuring superior stability. Furthermore, Bilinear Koopman achieves an RTF near 48—five times faster than the PINN. This efficiency highlights the advantage of a linear-in-the-state structure for real-time optimization over the iterative evaluation of deep neural layers.

VI. CONCLUSION

This study evaluated Bilinear Koopman and PINN architectures for the predictive control of a nonlinear quadruple tank system. Results demonstrate that the Bilinear Koopman model achieves superior identification and control performance by exploiting the system's control-affine structure. While both architectures reach VAF values above 99.0%, the Koopman framework effectively filters stochastic disturbances, resulting in significantly more stable and faster stabilization in closed-loop scenarios than the PINN, which exhibited higher sensitivity to sensor noise.

In the NMPC stage, the Koopman-based controller consistently outperformed the PINN approach, achieving a sinusoidal tracking RMSE of 0.0961 cm and a 44.8% reduction in global SSE during step response transients. The integration with CasADi enabled efficient real-time execution, where the bilinear structure favored rapid QP solutions via OSQP, whereas the PINN required more intensive SQP optimization. While PINNs offer high fitting potential, the structural robustness of Bilinear Koopman realizations provides a more reliable foundation for the predictive control of complex hydraulic processes. Future work will explore the adaptive scaling of lifting functions to maintain this fidelity under broader operational regimes.

DATA AND CODE AVAILABILITY

The source code, simulation results, and models used in this study are available to support reproducibility. These resources include the implementation of the PINN, Bilinear Koopman Realization, and NMPC architectures. Access the repository at: `https://github.com/azvcud/CoDiT_Comparison-of-PINN-and-Bilinear-Koopman`.

REFERENCES

[1] A. Annaswamy, K. H. Johansson, and G. J. P., eds., "Control for Societal-Scale Challenges: Road Map 2030," IEEE Control Systems Society, 2023.

[2] D. Santiago Aponte and D. Tellez-Castro, "Koopman-Based MPC for the Quadruple Tank System: An Experimental Comparison," in *2025 IEEE 7th Colombian Conference on Automatic Control (CCAC)*, pp. 1–6, 2025.

[3] M. Korda and I. Mezić, "Linear predictors for nonlinear dynamical systems: Koopman operator meets model predictive control," *Automatica*, vol. 93, pp. 149–160, 2018.

[4] R. Patel, S. Bhartiya, and R. Gudi, "Model Predictive Control using Physics Informed Neural Networks for Process Systems," *IFAC-PapersOnLine*, vol. 58, no. 14, pp. 775–780, 2024.

[5] B. O. Koopman, "Hamiltonian systems and transformation in Hilbert space," *Proc. Natl. Acad. Sci. U.S.A.*, vol. 17, no. 5, 1931.

[6] S. L. Brunton and J. N. Kutz, *Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control*. Cambridge, UK: Cambridge University Press, 2019.

[7] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[8] K. H. Johansson, "The quadruple-tank process: A multivariable laboratory process with an adjustable zero," *IEEE Trans. Control Syst. Technol.*, vol. 8, no. 3, pp. 456–465, 2000.

[9] M. K. Khaldi, M. Al-Dhaifallah, I. Aljamaan, F. M. Al-Sunni, O. Taha, and A. Alharbi, "From Block-Oriented Models to the Koopman Operator: A Comprehensive Review on Data-Driven Chemical Reactor Modeling," *Mathematics*, vol. 13, no. 15, Art. no. 2411, 2025.

[10] H. El-Hussieny, "Trajectory Tracking and Stabilization of Quadrotors Using Deep Koopman Model Predictive Control," *arXiv preprint arXiv:2508.13795*, 2025.

[11] T.-L. Habich, A. Mohammad, S. F. G. Ehlers, M. Bensch, T. Seel, and M. Schappler, "Generalizable and Fast Surrogates: Model Predictive Control of Articulated Soft Robots Using Physics-Informed Neural Networks," *IEEE Transactions on Robotics*, vol. 42, pp. 619–636, 2026.

[12] D. Chen, C.-K. Chui, and P. S. Lee, "Physics informed machine learning based predictive control for intelligent operation of edge datacenters," *Applied Energy*, vol. 402, Art. no. 126975, 2026.

[13] A. Taghieh, E. Cibaku, and S. Park, "Learning Koopman Observables via Kolmogorov-Arnold Networks for Power System Transient Analysis and Model Predictive Control," in *2025 IEEE 64th Conference on Decision and Control (CDC)*, pp. 387–394, 2025.

[14] M. Yan, M. Han, A. W.-K. Law, and X. Yin, "Self-tuning moving horizon estimation of nonlinear systems via physics-informed machine learning Koopman modeling," *AIChE Journal*, vol. 71, no. 2, Art. no. e18649, 2025.

[15] D. Bruder, X. Fu, and R. Vasudevan, "Advantages of Bilinear Koopman Realizations for the Modeling and Control of Systems with Unknown Dynamics," *CoRR*, vol. abs/2010.09961, 2020.